Cong Nguyen

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Game Systems Programmer with 2+ years of enterprise software engineering experience, specializing in Unity and C#. Architect of *Mo-Li*, a turn-based monster-collection RPG implementing an event-driven architecture, state-machine combat, and ScriptableObject-driven content pipelines. Combines disciplined Git workflows and production debugging from 47-member enterprise delivery with hands-on pixel art and combat UI design to bridge engineering and creative disciplines.

Technical Competencies

- **Game Dev & Architecture**: Unity Engine, C#, EventBus (struct events, replay cache), State Pattern, Command Pattern, Strategy Pattern, Object Pooling, ScriptableObjects, Addressables, Async/Await.
- **Game Systems & Data**: Turn-based combat engines, action queue priority handling, JSON save systems (ISaveable registry), polymorphic effect architectures (`[SerializeReference]`).
- **Creative & UI/UX**: Aseprite (pixel art, sprite sheets), combat UI/UX implementation (turn queue, status cards, responsive layouts), cross-functional collaboration with Artists and Designers.
- **Engineering Practices**: Git (feature/develop branch workflows, PR reviews), Docker, Postman, production debugging, systems integration, Agile/Scrum.
- **Other Technologies**: TypeScript, Angular, Node.js, Express.js, MongoDB, SQL, SAP ABAP.

Featured Game Project

- **Mo-Li (GitHub Repo)** — **Solo Developer**
  Turn-Based Monster-Collection RPG — *02/2026 – Present*
  - **Decoupled Game Architecture**: Architected core game sequence with an asynchronous Bootstrapper, persistent singleton managers (Audio, Input, Scene, Save, Assets), and strict World vs. Combat scene boundaries.
  - **Event-Driven Communication**: Engineered a generic `EventBus<T>` utilizing struct events and last-value replay caching, completely isolating gameplay simulation from UI updates and audio playback.
  - **Combat Engine & Action Queue**: Implemented turn-based combat with State Pattern sub-states for player decision flow (Action → Target → Confirm). Modeled skills and item usage as Command objects executed via an asynchronous `ActionQueue` with priority insertion for interrupt reactions (e.g., Faint) and reentrancy guards.
  - **Designer Content Pipeline**: Designed a data-driven system with ScriptableObjects loaded via Addressables; built a polymorphic effect pipeline (`[SerializeReference]` + Strategy Pattern) and bitwise flag target types (`[Flags]`), empowering designers to configure modular abilities without code modifications.
  - **Persistence & Memory Optimization**: Developed a 3-slot JSON save architecture with an `ISaveable` registry and a generic object pool (`IPoolable`) for dynamic combat UI elements and floating damage indicators.
  - **Art & UI/UX**: Hand-crafted original pixel art sprites and animations in Aseprite; designed responsive combat UI featuring dynamic turn-order timeline, unit cards, and cooldown/MP state visualization.

Professional Experience

- **FPT Software** — **Ho Chi Minh City, Vietnam**
  Software Engineer – SAP Technical Consultant — *02/2023 – 04/2025*
  - Collaborated within a 47-member cross-functional project team migrating a mission-critical enterprise system, gaining deep exposure to complex system architectures and structured Agile delivery.
  - Designed and implemented interface integrations between multiple distributed platforms (SAP PI/PO, IDOC), mastering robust data flow protocols, serialization, and error recovery—principles directly applicable to game networking and save architectures.
  - Engineered custom business logic and extensions, practicing structured problem-solving and rigorous production debugging in high-availability enterprise environments.
  - Developed and tested UI/UX components across enterprise platforms, ensuring cross-system responsiveness and usability.
  - Maintained disciplined Git version control, participating in feature-branch workflows, comprehensive code reviews, and cross-team alignment meetings.
- **FPT Software** — **Ho Chi Minh City, Vietnam**
  Angular Developer — *08/2022 – 02/2023*
  - Developed and maintained responsive web applications using the Angular framework and TypeScript.
  - Implemented clean component hierarchies, routing, state navigation, and integrated RESTful backend APIs.
  - Conducted unit testing, issue tracking, and collaborative sprint planning in an Agile environment.

Additional Technical Projects

- **E-Commerce System with Custom Recommendation Engine** — **UTC2**
  Graduation Capstone Project — *08/2023 – 02/2024*
  - Designed and built a Neighborhood-Based Collaborative Filtering recommendation model completely from scratch in TypeScript without relying on external ML libraries, demonstrating strong mathematical and algorithmic problem-solving.
  - Developed the full-stack e-commerce web platform (Angular, Express.js, MongoDB, TypeScript) handling authentication, cart management, and real-time personalized recommendations.

Education & Credentials

- **University of Transport and Communications Campus in HCMC (UTC2)** — **Ho Chi Minh City**
  Bachelor of Information Technology — *2019 – 03/2024*
  Core topics: Software Engineering, Object-Oriented Programming, Data Structures & Algorithms, Database Systems.

Languages

- **Vietnamese**: Native mother tongue.
- **English**: B1 (Professional working proficiency – comfortable reading technical documentation, writing clean code comments, and collaborating in English-speaking workplaces).